

Day 8 – Async Webhooks with Celery

Setup & Prerequisites

1. Setup Checklist:

- **Docker Running:** Ensure you have Docker running on your system.
- **Start Redis:** Run the Redis server in a container (`docker run -d -p 6379:6379 redis:7-alpine`).
- **Browser Open:** Open a browser window ready to access the Django Admin panel.
- **Terminal Tabs:** Prepare two separate terminal tabs: one for running the management command, and one for running the Celery worker daemon.

2. Prerequisites:

- You must have completed Day 7 successfully (Synchronous webhook sending working with `requests.post`).

Learning Objectives

By the end of this class, you will be able to:

- Explain the role of a message broker (Redis) and a task runner (Celery) in async applications.
- Configure Celery inside a Django project environment.
- Create a `WebhookDeliveryLog` database audit trail.
- Build resilient asynchronous Celery tasks using automatic retries and exponential backoff calculations.
- Use `.delay()` to dispatch tasks asynchronously.
- Run and monitor Celery worker processes from the command line.

Classroom Timeline (45 min)

Segment	Duration	Focus
1. Hook & Big Picture	7 min	Define background worker models. Map the producer-broker-consumer relationship.
2. Key Concepts & Core Ideas	7 min	Explain shared tasks, bindings, exponential backoff, and late acknowledgments.

3. Live Coding Walkthrough	23 min	Start Redis, configure Celery settings, write the Audit Log model, implement the Celery task, update the CLI command, and run the worker.
4. Check for Understanding	5 min	Q&A on Redis message queue states, worker crashes, and task acknowledgment flags.
5. Class Wrap-up & Checkpoint	3 min	Verify async webhook tasks execute in the worker terminal and write logs to admin.

Lecture Step-by-Step Delivery Plan

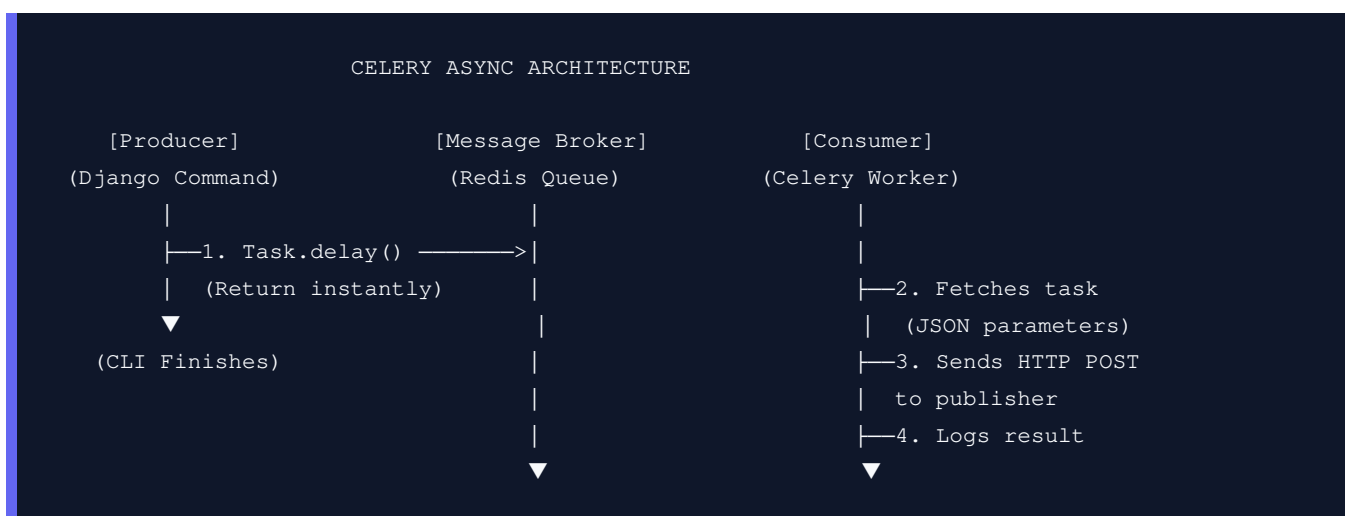
1. Hook & Big Picture (7 min)

What happens if a publisher's server goes down during our key check? In yesterday's code, the management command waited, blocked, and if the network request crashed, that webhook was lost forever. Today, we will throw the webhook job into a Redis buffer queue, let our management command finish instantly, and have a separate background process handle the HTTP delivery and automatic retries in the background.

- **Q&A:**

- **Question:** What happens if a publisher's server goes down during our key check?
 - **Answer:** In a synchronous system, the command blocks, taking several seconds per request. If multiple servers are down, the script hangs or crashes, and failed webhooks are lost forever.

- **Decoupled Architecture:** By isolating the execution queue, if our web server crashes, tasks in Redis remain safe. If a publisher is offline, Celery will wait and try again later.



2. Key Concepts & Core Ideas (7 min)

Introduce these asynchronous concepts:

- **Task Queue / Broker:** An intermediary (Redis) that receives messages from the publisher and buffers them until a consumer (Celery worker) retrieves them.
 - **@shared_task:** A decorator that defines a reusable Celery task without requiring explicit imports of the project-specific Celery application instance.
 - **bind=True:** Binds the task function to the task instance, passing the task object itself as `self` (enabling us to invoke retries).
 - **Exponential Backoff:** Instead of retrying immediately (when the remote server is likely still down), we double the wait time on each retry: $60 * (2 ** attempt)$ (i.e. 60s, 120s, 240s).
 - **Late Acknowledgment (`task_acks_late=True`):** Enforces that the worker only tells Redis it completed the task *after* the function successfully executes. If the worker crashes mid-task, Redis re-queues the task for another worker.
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3. Live Coding Walkthrough (23 min)

Step 3.1: Starting the Redis Message Broker

- **Concept:** Celery needs a queue to store tasks. We will spin up a lightweight Redis instance in a background Docker container.
- **Code:**

```
docker run -d -p 6379:6379 redis:7-alpine
```

- **Verify:** Verify that Redis is listening by running `docker ps` or connecting via `redis-cli ping`.

Step 3.2: Configuring Celery inside Django

- **Concept:** Create the Celery configuration file in the settings directory, update package initialization files, and add broker URLs to Django settings.
- **Code:** Create `gamekey_platform/celery.py`:

```
import os
from celery import Celery

# Set default settings module
os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'gamekey_platform.settings')

app = Celery('gamekey_platform')
# Reads configurations from settings.py starting with CELERY_ prefix
app.config_from_object('django.conf:settings', namespace='CELERY')
# Discover tasks.py automatically inside apps folders
app.autodiscover_tasks()
```

Update `gamekey_platform/__init__.py` to load Celery on startup:

```
from .celery import app as celery_app

__all__ = ('celery_app',)
```

Add these configurations to the bottom of `gamekey_platform/settings.py`:

```
CELERY_BROKER_URL = 'redis://localhost:6379/0' # Path to Redis DB 0
CELERY_RESULT_BACKEND = 'redis://localhost:6379/0'
CELERY_TASK_ALWAYS_EAGER = False # set True in tests to run tasks synchronously
```

- **Verify:** Run `celery -A gamekey_platform inspect ping` to ensure Celery can connect to Redis.

Step 3.3: Creating the Webhook Audit Log Model

- **Concept:** In production, we must keep a permanent log of all webhook notifications, retries, and delivery responses for auditing and debugging.
- **Q&A:**
 - **Question:** Why do we log every delivery attempt (success or failure)?
 - **Answer:** It provides an audit trail to debug delivery failures, prove webhooks were sent, and trace response status codes.
- **Code:** Add `WebhookDeliveryLog` to `games/models.py`:

```
class WebhookDeliveryLog(models.Model):
    game_key = models.CharField(max_length=50)
    publisher = models.ForeignKey(Publisher, on_delete=models.CASCADE)
    payload = models.JSONField()
    response_status = models.IntegerField(null=True, blank=True)
    error_message = models.TextField(blank=True)
    attempt = models.IntegerField(default=0)
    success = models.BooleanField(default=False)
    created_at = models.DateTimeField(auto_now_add=True)

    class Meta:
        ordering = ['-created_at']

    def __str__(self):
        status = 'OK' if self.success else 'FAIL'
        return f"[{status}] {self.game_key} attempt #{self.attempt}"
```

Run the database migrations and register the model:

```
python manage.py makemigrations
python manage.py migrate
```

Register the model in `games/admin.py`:

```
from .models import WebhookDeliveryLog
admin.site.register(WebhookDeliveryLog)
```

- **Verify:** Log into Django Admin and verify that the `Webhook Delivery Logs` section is visible.

Step 3.4: Writing the Celery Task with Retries

- **Concept:** Create `games/tasks.py`. This task retrieves the publisher settings, builds the payload containing the current attempt count, signs it, and dispatches the POST request. If the HTTP call fails or raises an error, the task registers a failure log and schedules a retry with exponential backoff.
- **Code:** Create `games/tasks.py`:

```
import hmac
import hashlib
import json
import requests
from celery import shared_task
from .models import Publisher, WebhookDeliveryLog

# bind=True binds the task instance to 'self' so we can invoke self.retry()
# max_retries limits the retry count.
@shared_task(bind=True, max_retries=3, default_retry_delay=60)
def send_expiry_webhook_async(self, publisher_id, game_key_str, game_title, expired_at_iso, attempt):
    publisher = None
    payload = {}

    try:
        publisher = Publisher.objects.get(id=publisher_id)

        payload = {
            "event": "game_key.expired",
            "game_key": game_key_str,
            "game_title": game_title,
            "expired_at": expired_at_iso,
            "attempt": attempt,
        }

        secret = publisher.webhook_secret.encode('utf-8')
        body = json.dumps(payload, sort_keys=True).encode('utf-8')
        signature = hmac.new(secret, body, hashlib.sha256).hexdigest()

        headers = {
            "Content-Type": "application/json",
            "X-Signature": f"sha256={signature}",
        }

        response = requests.post(
            publisher.webhook_url,
            json=payload,
            headers=headers,
```

```

        timeout=5,
    )

    # Log attempt result in audit table
    WebhookDeliveryLog.objects.create(
        game_key=game_key_str,
        publisher=publisher,
        payload=payload,
        response_status=response.status_code,
        attempt=attempt,
        success=response.status_code < 400,
    )

    if response.status_code >= 400:
        raise Exception(f"HTTP {response.status_code} from publisher endpoint")

except Exception as exc:
    if publisher:
        # Log the error details
        WebhookDeliveryLog.objects.create(
            game_key=game_key_str,
            publisher=publisher,
            payload=payload,
            error_message=str(exc),
            attempt=attempt,
            success=False,
        )

    # Exponential backoff countdown logic:
    # attempt 0 -> countdown = 60 * 2^0 = 60 seconds
    # attempt 1 -> countdown = 60 * 2^1 = 120 seconds
    # attempt 2 -> countdown = 60 * 2^2 = 240 seconds
    raise self.retry(exc=exc, countdown=60 * (2 ** attempt))

```

⚠ Common Pitfalls:

- **Omitting `raise on retry`:** If you call `self.retry()` without raising it, the function execution doesn't stop, which can result in double logs. Always write `raise self.retry()`.

Step 3.5: Updating the Management Command

- **Concept:** Update `check_expired_keys.py` to dispatch tasks to the queue using `.delay()` instead of calling the function synchronously. Note that because we update the status field in bulk via `.update()`, we must materialize the queryset to a list *before* running the update.
- **Code:** Update `games/management/commands/check_expired_keys.py`:

```

from django.core.management.base import BaseCommand
from django.utils import timezone
from games.models import GameKey
from games.tasks import send_expiry_webhook_async # Import async task

```

```

class Command(BaseCommand):
    help = 'Mark expired keys and dispatch async webhook notifications.'

    def handle(self, *args, **options):
        expired_keys = GameKey.objects.select_related('game__publisher').filter(
            expires_at__lte=timezone.now(),
            status='active'
        )

        # Materialize into a list because bulk update() clears queryset references
        keys_to_notify = list(expired_keys)
        count = expired_keys.update(status='expired')

        # Loop and dispatch Celery tasks
        for key in keys_to_notify:
            # .delay() pushes the execution parameters into Redis. Celery worker handles the r
            send_expiry_webhook_async.delay(
                publisher_id=key.game.publisher.id,
                game_key_str=key.key_string,
                game_title=key.game.title,
                expired_at_iso=key.expires_at.isoformat(),
                attempt=0,
            )

        self.stdout.write(self.style.SUCCESS(
            f'Expired {count} keys. Webhook tasks dispatched to Celery.'
        ))

```

Step 3.6: Running the Asynchronous Stack

- **Concept:** Run the Celery worker process and trigger the expiration flow to watch the system in action.
- **Code:** In terminal tab 1, start the background worker:

```
celery -A gamekey_platform worker --loglevel=info
```

In terminal tab 2, trigger the expiration check command:

```
python manage.py check_expired_keys
```

- **Verify:** Watch the logs in terminal tab 1. You should see the tasks received, logged, and executed asynchronously. Open Django admin and verify that corresponding `WebhookDeliveryLog` rows are created.

4. Check for Understanding (5 min)

Question: "What's in Redis right now, after `.delay()` is called but before the worker picks up the task?"

Answer: Redis contains a serialized JSON message (the task payload) representing the Celery task. This message includes the task name, task ID, arguments (like `publisher_id`, `game_key_str`), and other metadata, waiting in a list (acting as a queue).

Question: "What happens if the Celery worker crashes mid-task? Is the task lost?"

Answer: With default settings (late acknowledgment disabled), the task is acknowledged when the worker starts, meaning it could be lost if it crashes mid-execution. If late acknowledgment is enabled (`task_acks_late = True`), the task is only acknowledged after successful execution; if the worker crashes, the broker (Redis) will re-queue the task for another worker.

Question: "Why do we log both successful and failed deliveries to `WebhookDeliveryLog`?"

Answer: Logging both creates a complete audit trail. It allows developers to diagnose issues, prove to publishers that webhooks were dispatched, analyze error patterns, and keep track of execution history for debugging delivery problems.

5. Daily Checkpoint & Wrap-up (3 min)

- **Verification Criteria:**

- ☐ The Celery background worker starts cleanly.
- ☐ Running `check_expired_keys` runs instantly without blocking for HTTP calls.
- ☐ The Celery worker console outputs task receipt and execution details.
- ☐ The `WebhookDeliveryLog` entries populate in the admin panel showing accurate response statuses and attempt numbers.